

Matlab

% M file: To convert state-space of a system into transfer function form.

% Defining the state space system

A = [-5 -6 -2; 1 0 0; 0 1 0];

B = [1; 0; 0];

C = [0 0 10];

D = [0];

disp('First transfer function=')

[num,den]= ss2tf(A,B,C,D)

sys_tf=tf(num,den)

disp(sys_tf)

Command Window Output

First transfer function=

num =

0 0 0 10

den =

1.0000 5.0000 6.0000 2.0000

sys_tf =

 10

s^3 + 5 s^2 + 6 s + 2

Continuous-time transfer function.